# Visual-Document RAG | Why This Design, and What the Evidence Says

**Operation Autopilot design exercise · Akshat Bajpai**

**Repo** github.com/Akbonline/visual-document-rag
**Corpus** vidore/vidore_v3_hr @ 0cdf0979
1,110 pages · 318 English queries · 23 Aug 2026

**Position.** I would ship an inspectable OCR-first system, measure its failures, and add visual retrieval only where evidence justifies the cost. The prototype tests those decisions, not a homegrown vector index. Four iterations narrowed the solution: V1 framed the system; V2 made dataset boundaries reusable; V3 converted research into contracts and separated industrial design from weekend scope; V4 added live generation and measured failures. **Each version removed unsupported assumptions rather than adding architecture for presentation.**

| 0.5194 | 41 ms | 3.19 s | $2.63 | 48 |
|---|---|---|---|---|
| NDCG@10 · HYBRID · 318 Q | RETRIEVAL P95 (BUDGET 300) | TTR P95 (BUDGET 6 S) | PER 1,000 ANSWERED QUERIES | MODEL CALLS · 24 PAIRS · 0 PROVIDER FAILURES |

## 1 · INGESTION | OCR FIRST, VISUAL AS A MEASURED ARM

*Decision: build the transparent path, quantify what it costs, add the expensive path only where the cost is repaid.*

**Why.** OCR makes the baseline inspectable: operators trace exact text, identifiers, and coordinates, while the system gets lexical retrieval and quote-level citations on CPU. The deciding evidence is generalization. On the out-of-domain DocDeg benchmark, OCR plus Llama scores **0.523** nDCG@5 versus ColPali at **0.274**; ViDoRe V3 reports the strongest full pipeline as text-only, **63.6** versus **57.8** nDCG@10 for visual. Visual retrieval's in-domain advantage is real, but Autopilot's proprietary robot documentation is outside that benchmark distribution. It should earn its storage, hardware, and citation complexity against a measured baseline, not by assumption.

**What breaks in each.** OCR loses meaning carried only by layout, and token overlap understates that because a flattened table can score well while becoming unusable. Visual late interaction preserves layout but costs ~**30×** the storage of a dense text index, needs stronger hardware, and returns pages rather than quotable spans. **I measured the OCR tax: 3.0% nDCG@10** against publisher-supplied text (0.4865 → 0.4719), with token F1 **0.899**.

## 2 · CHUNKING | WHY I THREW MY OWN RESULT AWAY

*Decision: fixed 180-token windows ship; structure-aware was built, tested, and rejected.*

**Why it matters more than the result.** A boundary through a table is the failure the question is really about, so I built the structure-aware policy under identical candidates and token budgets, then ran a **repeated oracle arm with byte-identical prompts** as a control.

**FINDING** Retrieval got *worse* (0.4742 vs 0.4865). Generation looked better: retrieved arm **+0.0123** F1, but the control moved **+0.0148**, more than the treatment. **Noise-adjusted delta -0.0024.** Preserving block boundaries is insufficient without real layout recovery. Without the control I would have shipped a false win.

## 3 · RETRIEVAL | THE BUDGET I SET FIRST

*Decision: sparse + dense in parallel, projected chunk→page, fused by RRF. No reranker.*

**Why RRF, not score blending.** BM25 and cosine live on incompatible scales; fusing *ranks* needs no calibration, and calibration would consume a tuning split from only 318 queries. **Why no reranker:** I required a measured ranking failure before spending the latency. I now have one in §5, which is how the decision should unlock.

Sparse leads dense here (0.4865 vs 0.4400): this corpus is procedural text full of exact identifiers, BM25's home ground. Fusion beats both: **+6.8%** nDCG@10, **+8.8%** Recall@10. **Budget stated in advance and met:** retrieval <300 ms p95 (measured **41.3 ms**), response <6 s p95 (measured **3.19 s**).

## 4 · EVALUATION | THE PART I WOULD DEFEND HARDEST

*Decision: metrics are gated by a machine-readable capability contract; retrieval and generation are separated by an oracle arm.*

**Why gate metrics.** One binary gold per query makes Recall, MRR, and nDCG the same number wearing three hats. Each adapter must therefore declare judgment unit, cardinality, scale, grades, threshold, AND/OR structure, and reference-answer availability. The runner enables only supported metrics and returns `not_applicable` for the rest. It never silently approximates.

**Why an oracle arm.** It is the cleanest way to ask whether retrieval or the model caused the miss. With identical prompts and budgets, only the pages differ: gold-page citation recall **0.249→0.544**, answer F1 **0.452→0.488**. Citation coverage more than doubles for only **+0.037** F1. Retrieval contributes materially, but better pages alone do not solve interpretation. **Both stages limit this corpus.**

**HONESTY** ViDoRe V3 declares multi-page judgment structure `unknown`. I report **19 of 24** Gate C queries as `unscorable_gold` for stage attribution rather than assuming `all_of` and manufacturing causes. All other metrics still reported.

## 5 · WHERE IT FALLS OVER | WHAT A WEEK BUYS

*Every item below is chosen from a measurement, not from novelty.*

- **Ranking and coverage both limit.** Recall@1 **0.186** vs Recall@10 **0.579**: some gold is ranked badly, while 42% never enters the top 10 against ~5.4 gold pages/query. **→ test a cross-encoder reranker for the first and multi-gold-aware retrieval for the second.**
- **Tables are the weakest evidence type, not charts.** nDCG@10 **0.443** vs 0.523 for text. **→ table structure recovery before visual expansion.**
- **Infographics have the lowest recall** (**0.396**) despite the highest nDCG: when found they rank well, but much labeled evidence is missed. **→ this is where a visual arm earns its experiment.**
- **Oracle F1 remains 0.488.** Better pages do not fix visual interpretation. **→ test multimodal generation after improving retrieval.**

## 6 · NATIVE .DOCX / .PPTX

**Why the benchmark misleads here.** ViDoRe supplies rendered images and forces structure reconstruction; real native files already contain heading hierarchy, table cell grids, slide coordinates, and speaker notes. OCR over a rendered `.docx` discards available structure, then pays to rebuild it worse. My 3.0% OCR tax is the floor on that.

Two cases: **speaker notes are load-bearing and invisible in any render**; **PowerPoint has no pagination until you render it**, so "page 42" is an artifact of my renderer while a slide shape ID is verifiable in the customer's own file. The qualifier: native structure is rich only where the format carries *semantics*. PDF positions glyphs by coordinate, which is why native PDF streams mangle multi-column order and math. **Parse native where semantic, render for fidelity and citation, OCR as fallback.**

## 7 · COST, TOKENS, TTR, AND THE HUMAN IN THE LOOP

**COST** Measured: **$0.00263**/retrieved-context answer → **$2.63** per 1,000; the full 318-query paired experiment projects to **$1.57**. OCR/indexing run locally, with no per-call model charge.

**TOKENS** 2,276 in / 206 out per query. Prompt caching cut the oracle arm **-35.4%** ($0.0551→$0.0356) on 30,464 cached tokens. A 256-token output ceiling truncated structured JSON mid-citation: **a ceiling below the safe range wastes the call and makes retries more expensive than one adequate call.** I measure the output-token distribution and retry only genuine `max_output_tokens` failures.

**TTR** Prompt submission to completed response, decomposed per stage: p50 **1.51 s**, p95 **3.19 s**. Smaller output ceilings do *not* necessarily improve TTR because truncation plus retry inflates the tail.

**HRI** A correctness constraint, not decoration. An operator assistant must return a complete cited answer, an explicit refusal, or a typed failure. Truncated JSON is not usable interaction. **7 of 24** retrieved-context calls returned explicit refusals rather than unsupported answers. Voice transcription can mangle identifiers ("ERR-42" → "error forty two"), weakening BM25's exact-match advantage. The operational metric is **time-to-verified-action**, not response speed: show answer, page, quote, and crop together so a technician can verify before acting.

**Scope boundary.** Not built: ColPali, learned reranking, native Office ingestion, a second benchmark adapter, or production deployment; each is reasoned about in the repository. Not claimed: visual localization from a text-only path or full any-of/all-of attribution. Full measured evidence is in the accompanying appendix; the 52-decision history is in `09_REASONING_LOG.md`. Sources for the generalization argument: DocDeg and ViDoRe V3.